BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, SAHIWAL

INTERMEDIATE (11TH CLASS) EXAMINATION, 2021

MATHEMATICS

Past Paper 2021 • Paper Code: 6193

Retyped faithfully from the original board paper — same questions, no changes.

Mathematics - Paper: I - Objective (II)

Time Allowed: 30 Minutes | **Maximum Marks:** 20 | **Paper Code:** 6193

Note: Note: - You have four choices for each objective type question as A, B, C and D. The choice which you think is correct; fill that circle in front of that question number in your answer book. Use marker or pen to fill the circles. Cutting or filling up two or more circles will result no mark.

Note: Objective - (II)

Q1. If a,b,c have their usual meanings then $(c^2+a^2-b^2)/(2ac)$ =

(A) cos α
(B) cos β
(C) cos γ
(D) sin β

Q2. If n is any positive integer, then $2^n > 2(n+1)$ is true for all.

(A) $n \leq 3$
(B) $n < 3$
(C) $n \geq 3$
(D) $n > 3$

Q3. What angle is quadrantal angle?

(A) 120°
(B) 270°
(C) 60°
(D) 45°

Q4. If cos θ = 1/√2, then θ is equal to:

(A) 30°
(B) 45°
(C) 60°
(D) 90°

Q5. sin(3π/2 - θ) =

(A) sin θ
(B) cos θ
(C) -sin θ
(D) -cos θ

Q6. The period of 3sin 3x is:

(A) π/3 (B) π/2

(C) 2π/3 (D) π

Q7. ${}^{n}P_{r}/r!$ is equal to:

(A) [illegible in source scan] (B) ${}^{n}C_{r-1}$

(C) ${}^{n+1}C_{r}$ (D) ${}^{n-1}C_{r}$

Q8. Radius of the inscribed circle is:

(A) r = Δ/S (B) r = abc/(4Δ)

(C) r = S/Δ (D) r = (S-a)/Δ

Q9. Rank of matrix [1 0 0; 0 1 0; 0 0 0] is:

(A) 1 (B) 2

(C) 3 (D) 4

Q10. $(x^3+1)/((x-1)(x+2))$ is:

(A) proper fraction (B) improper fraction

(C) identity (D) both B & C

Q11. Any real number "a" is equal to:

(A) ia (B) (a,b)

(C) a (D) (b,a)

Q12. $\{x|x\in E \wedge 4<x<6\}$ equals:

(A) {4} (B) {5}

(C) {6} (D) φ

Q13. If A is a matrix of order 3×4, then the order of AA^t is:

(A) 4×3 (B) 3×3

(C) 3×4 (D) 4×4

Q14. $Tan^{-1}(-\sqrt{3})$ =

(A) 2π/3 (B) -2π/3

(C) -π/6 (D) -π/3

Q15. An equation of the form $ax^2+bx+c=0$ is called quadratic if:

(A) a=0 (B) b=0

(C) a≠0 (D) b≠0

Q16. If one root of the equation $x^2-3x+a=0$ is 2, then a is:

(A) 2 (B) -2

(C) [illegible in source scan] (D) -3

Q17. If sin x = -√3/2, then solution is:

(A) 4π/6, 5π/6 (B) 4π/3, 5π/3

(C) 5π/6, 7π/6 (D) π/3, 7π/3

Q18. If a_{n-3} = 2n-5, its nth term is:

(A) 2n+1 (B) 2n+3

(C) 2n-2 (D) 2n-8

Q19. If a,b are in A.P, then 2A is:

(A) a-b (B) (a+b)/2

(C) a+b (D) b-a

Q20. Let A,G,H be the A.M, G.M and H.M between a and b respectively, then G^2 =

(A) A+H (B) √(ab)

(C) A/H (D) AH

Mathematics - Paper: I - Subjective

Time Allowed: 2:30 Hours | **Maximum Marks:** 80

Note: Note: - Section I is compulsory. Attempt any three questions from section II.

Note: Section - I

Q2.. (Marks 16) Write short answers to any Eight parts. (8 x 2 = 16)

i. Find the multiplicative inverse of (√2, -√5)

ii. Prove that -7/12 - 5/18 = (-21-10)/36

iii. If z_1=2+i, z_2=3-2i, z_3=1+3i then express $(\bar{z}_1\bar{z}_2)/z_3$ in the form a+ib

iv. Write the inverse and contrapositive of conditional ~p→~q

v. Show A-B and B-A by Venn diagram when A and B are overlapping sets.

vi. If a and b are elements of a group G, then solve the equation xa=b

vii. Find the matrix X if X[5 2; -2 1] = [-1 5; 12 3]

viii. Show that |1 1 1; x y z; yz zx xy| = |1 1 1; x y z; x^2 y^2 z^2|

ix. If A=[-1 2; 1 4; 2 -1] and B=[0 3; 1 2] show that $(AB)^t = B^tA^t$

x. Find two consecutive numbers, whose product is 132.

xi. Evaluate $(-1+\sqrt{-3})^5 + (-1-\sqrt{-3})^5$

xii. Find numerical value of K, if the polynomial x^3+kx^2-7x+6 has a remainder of -4, when divided by x+2.

Q3.. (Marks 16) Write short answers to any Eight parts. (8 x 2 = 16)

i. Write into partial fraction form of $4x^2/((x^2+1)^2(x-1))$ without finding constants.

ii. Write into partial fraction form of $1/((x-1)^2(x^2+2))$ without finding constants.

iii. If $a_{n-3} = 3n-11$, find nth term of the sequence.

iv. Find the Geometric Mean between -2i and 8i.

v. If $y = 1+2x+4x^2+8x^3+...$ show that $x = (y-1)/(2y)$

vi. Find 8th term of H.P: 1/2, 1/5, 1/8,...

vii. Write (52.51.50.49)/(4.3.2.1) in the factorial form.

viii. Find the value of n when ${}^nP_r = 11.10.9$ [best-effort]

ix. Find the value of n, when ${}^nC_3 = {}^nC_4$

x. Show that the inequality $4^n > 3^n+4$ is true, for integral values of $n \geq 2$.

xi. Calculate $(9.98)^4$ by means of binomial theorem.

xii. Expand $(1+x)^{-1/3}$ upto 4 terms.

> Note: (2)

Q4.. (Marks 18) Write short answers to any Nine parts. (9 x 2 = 18)

i. Convert 54°45' into radians.

ii. Verify that $2\sin45° + 1/2\ \text{cosec}45° = 3/\sqrt{2}$

iii. Prove that $\sec^2\theta - \text{cosec}^2\theta = \tan^2\theta - \cot^2\theta$

iv. Prove that $\sin(180°+\alpha)\sin(90°-\alpha) = -\sin\alpha\cos\alpha$

v. Prove that $\cot\alpha - \tan\alpha = 2\cot2\alpha$

vi. Express cos32° - cos48° as product. [best-effort]

vii. Find the period of sin(x/3).

viii. Find θ, if sin θ = 0.5791

ix. Find the smallest angle of the triangle ABC, when a=37.34, b=3.24, c=35.06

x. Find the value of R if a=13, b=14, c=15

xi. Find the value of $\cos(\sin^{-1}(1/\sqrt{2}))$

xii. Solve the equation $1+\cos x=0$

xiii. Find the solution of $\sec x = -2$

> Note: Section - II (Each question carries (5+5=10) Marks)

Q5.. (a) Find the rank of the matrix [1 -4 -7; 2 -5 1; 1 -2 3; 3 -1 4] [best-effort] (b) Use synthetic division to find the values of a and b if -2 and 2 are the roots of the polynomial x^3+ax^2+bx+c [best-effort]

Q6.. (a) Resolve into partial fraction $x^2/((x-2)(x-1)^2)$ (b) The sum of an infinite geometric series is 9 and the sum of squares of its terms is 81/5. Find the series.

Q7.. (a) Two dice are thrown. What is the probability that the sum of the number of dots appearing on them is 4 or 6? (b) Find the coefficient of x^5 in the expansion of $(x^2 - 3/(2x))^{10}$

Q8.. (a) Find the values of the trigonometric functions of the angle $-17\pi/3$ (b) Prove that $\cos3\theta/\cos\theta + \sin3\theta/\sin\theta = 4\cos2\theta$

Q9.. (a) Prove that $1/r^2 + 1/r_1^2 + 1/r_2^2 + 1/r_3^2 = (a^2+b^2+c^2)/\Delta^2$ (b) Prove that $\tan^{-1}(120/119) = 2\cos^{-1}(12/13)$ [best-effort]